



Boron group element doping of $\text{Li}_{1.5}\text{Al}_{0.5}\text{Ge}_{1.5}(\text{PO}_4)_3$ based on microwave sintering

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Received: 5 July 2020 / Revised: 2 September 2020 / Accepted: 22 September 2020
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Abstract

We report effectiveness of dopants selected from group 13, such as B, Ga, and In, on the conductivity of $\text{Li}_{1.5}\text{Al}_{0.5}\text{Ge}_{1.5}(\text{PO}_4)_3$ (LAGP) that is recognized as one of the promising solid electrolytes for next-generation solid-state lithium batteries. To minimize loss of versatile element, crystallization of LAGP using a microwave irradiation is introduced. Experimental results demonstrate that B enhances grain boundary conductivity, while Ga improves the bulk ionic conductivity. Co-doping of B and Ga leads to enhanced total ionic conductivity and remarkably reduced activation energy.

Keywords Lithium-ion batteries · Solid electrolyte · NASICON · Boron group

Introduction

Lithium-ion batteries (LIBs) have been widely used in portable electronic devices, such as mobile phones, laptops, and cameras, and in large-scale applications, such as electric vehicles [1, 2]. Although LIBs have been widely used, they are still suffering from safety concerns and electrochemical stability issues due to the use of organic electrolytes which have narrow electrochemical stability windows and high flammability. To address these issues, the liquid organic electrolytes need to be replaced by solid-state non-flammable Li^+ super ionic conductors [3–7].

Several Li^+ super ionic conductors have been reported and studied in the past decades of years, such as amorphous

$\text{Li}_x\text{PO}_y\text{N}_z$ (LiPON), cubic-garnet-structured $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ (LLZO), NASICON (Na^+ ion super conductor)-structured $\text{Li}_{1+x}\text{Al}_x\text{Ge}_{2-x}(\text{PO}_4)_3$ (LAGP), etc. [8–17] For the advantages of high ionic conductivity and good electrochemical stability, NASICON electrolytes have attracted a lot of attentions. In the NASICON structure, $\text{LiM}_2(\text{PO}_4)_3$ ($M = \text{Ti}, \text{Ge}, \text{Zr}$), the NASICON skeleton consists of MO_6 octahedra and PO_4 tetrahedra which form a three-dimensional network by corner-sharing. In the structure (space group $R\bar{3}C$), Li^+ ions occupy 6b and 36f sites, and M, P, and O ions are placed in 12c, 18e, and 36f sites, respectively. The concentration of Li^+ ion and the dimension of bottle neck of the Li^+ transporting path are the main factors that control Li^+ ionic conductivity [18]. Doping with trivalent ions such as Al, Cr, Ga, etc., into NASICON structure ($\text{Li}_{1+x}\text{M}_x\text{Ge}_{2-x}(\text{PO}_4)_3$), can introduce excess Li^+ ions due to charge compensation and adjust bottle neck size for Li^+ ion transportation at the same time [19–22]. Among them, Al-doped $\text{Li}_{1+x}\text{Al}_x\text{Ge}_{2-x}(\text{PO}_4)_3$ (LAGP) with $x = 0.5$ has been proved a promising electrolyte with a Li^+ ionic conductivity of about 10^{-4} S/cm. However, the ionic conductivity still cannot meet the practical requirements and need to be improved. Furthermore, its activation energy and the physical properties (mechanical strength, formation of voids and cracks, etc.) also need to be optimized through adjusting the composition and structure and preparing the processes [23–26].

In the traditional crystallization process of the melting-quench-processed LAGP [27], muffle furnace or tube furnace has been mainly used to crystallize glass precursor for LAGP. Heat transfers from the surface to inside the glass occur by

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Electronic supplementary material The online version of this article (<https://doi.org/10.1007/s10008-020-04829-2>) contains supplementary material, which is available to authorized users.

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heat conduction, which often causes deformation, cracks, and uneven crystallization due to uneven heat distribution. Long heating duration (a few to 10 h) may also lead to loss of some volatile element, such as Li, which lowers Li^+ ion concentration, resulting in reducing the Li^+ ionic conductivity.

In the microwave-assisted crystallization, the microwave energy is absorbed by the materials and converts to the kinetic energy of molecules. The heat is generated almost simultaneously in whole samples and temperature gradient can be ignored [26]. We used the microwave-assisted crystallization to crystallize $\text{Li}_{1.5}\text{Al}_{0.5-x}\text{Ga}_x\text{Ge}_{1.5}(\text{PO}_4)_3$ [27]. The duration of crystallization was drastically reduced to 30 min, and no microcrack was observed in the crystallized sample. The Ga^{3+} substitution influenced lattice parameter (bottleneck size), resulting in Li^+ ionic conductivity. The highest bulk Li^+ ionic conductivity of $6.67 \times 10^{-4} \text{ S cm}^{-1}$ was obtained at $x = 0.1$.

In this work, the boron group elements were used as a substituent for Al in LAGP to further increase in the ionic conductivity due to change of the bottleneck size. The substituted LAGP was crystallized by the microwave-assisted crystallization, and the Li^+ ionic conductivity and other properties were characterized.

Experimental

Synthesis of the glass ceramic

The $\text{Li}_{1.5}\text{Al}_{0.5-x}\beta_x\text{Ge}_{1.5}(\text{PO}_4)_3$ ($\beta = \text{B, Ga, In}$, $x = 0.1\sim 0.5$, denoted as LA β GP) glass ceramics was prepared by a combination of the melt-quenching with the microwave-assisted crystallization. The stoichiometric amount of GeO_2 (Aladdin, 99.999%), Li_2CO_3 (Aladdin, 99.99%), Al_2O_3 (Aladdin, AR), B_2O_3 (MK, AR), Ga_2O_3 (Aladdin, AR), In_2O_3 (Aladdin, AR), and $(\text{NH}_4)_2\text{H}_2\text{PO}_4$ (SCR, AR) was weighted. All reagents were purchased from Aladdin and used without further purification prior to use. The precursors were mixed by a ball-milling machine (Retsch PM100) at 200 rpm for 2 h. The ball-milled powder was grinded in an agate mortar for 10 min followed by a heat treatment at 380 °C for 6 h to decompose ammonia, after which, the powder was grinded for 10 min again followed by melting in a Pt crucible at 1380 °C for 2 h in a muffle furnace. The transparent, homogeneous, and viscous melt was quenched by pouring into a preheated mold (400 °C) with an inner diameter of 12 mm. The mold was then transferred into a muffle furnace at 400 °C and held for 1 h to release the thermal stresses and then cooled to room temperature in the furnace. The glass cylinder was cut into glass disks with a thickness of 250 μm using a diamond wire saw (STX-202A). The $\text{Li}_{1.5}\text{Al}_{0.5-x}\beta_x\text{Ge}_{1.5}(\text{PO}_4)_3$ disk was then put onto a double-layered plate that was stacked by quartz sheet and SiC sheet. LABGP/quartz/SiC was

crystallized at 900 °C in a microwave tube furnace (HY-Z61576) for 30 min. The SiC sheet can absorb microwave at low temperature and preheat LA β GP through heat conduction. Quartz was used to prevent the reaction between LA β GP and SiC during heating.

Assembly of full all-solid-state Li-ion battery using LA β GP solid electrolyte

Poly (acrylic acid) (PAA) (molecular weight 450,000, Aladdin) was dissolved into deionized water and magnetically stirred for 12 h to obtain 2 wt% PAA solution. LiOH (99.0%, Aladdin) was dropped into PAA solution according to the molar ratio of 1:1, and then the solution was further magnetically stirred for 12 h. Finally, PAA with Li-ion solution (PAALi) was obtained.

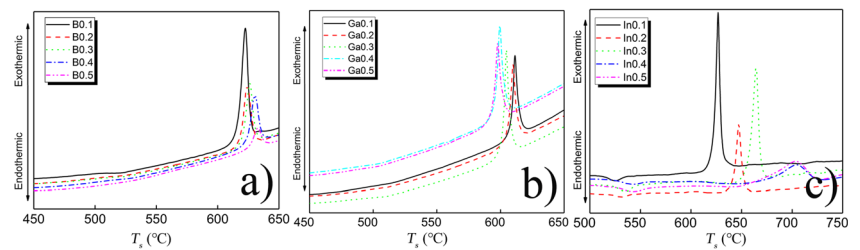
To prepare the cathode and anode electrode slurries, commercial $\text{LiNi}_{0.5}\text{Mn}_{1.5}\text{O}_4$ (99%, Sigma-Aldrich) and RuO_2 (99.9%, Aldrich) were respectively ball-milled with carbon nanotubes (CNT) (diameter 10–20 nm, length 5–15 μm , Shenzhen Nanotech. Port. Co. Ltd.) and the PAALi solution at a weight ratio of 90:2:8 by a ball-milling machine at 120 rpm for 2 h. The mixture was generally viscous that were then immediately used for assembling full all-solid-state Li-ion batteries.

The dried solid electrolyte $\text{Li}_{1.5}\text{Al}_{0.4}\text{Ga}_{0.1}\text{Ge}_{1.5}(\text{PO}_4)_3$ was weighed and coated with positive electrode slurry on one side. The cell was dried in an oven at 110 °C for 30 min. The mass of positive electrode was then weighed. The same process was also used to determine the anode loading. The assembled all-solid-state battery was then dried in a drying box at 80 °C for 12 h. After drying, both sides of the all-solid-state batteries were sputtered with Au to ensure electrical contact with current collectors (aluminum and copper foils for cathode and anode, respectively). To avoid the absorption of moisture from air during the electrochemical measurement, the all-solid-state batteries were wrapped with Parafilm (PM996, Bemis).

Characterization

The thermal behavior of the glass (glass transition temperature, T_g , crystallization temperature, T_c , and melting temperature, T_m) was analyzed by differential scanning calorimetry (DSC, Mettler Toledo TGA/DSC3+, Switzerland) from room temperature to 1000 °C at a heating rate of 10 °C/min. The crystal structure of the specimens was analyzed by X-ray diffraction (XRD, Smar/SmartLa, Japan) with $\text{Cu K}\alpha$ radiation in 2θ range from 10 to 80° at a scan rate of 2°·min⁻¹. The density of the samples was measured by Micrometrics, AccuPyc II 1340. The surface morphology of the crystallized specimens was investigated by scanning electron microscopy (SEM, Phenom-World BV, American). The Li-ion conductivity of the LA β GP disk

Fig. 1 DSC thermograms of $\text{Li}_{1.5}\text{Al}_{0.5-x}\beta_x\text{Ge}_{1.5}(\text{PO}_4)_3$ ($x = 0.1\sim 0.5$): (a) $\beta = \text{B}$, (b) $\beta = \text{Ga}$, and (c) $\beta = \text{In}$



was measured using the AC impedance method. Au was sputtered onto the two faces of the LA β GP disk to prepare the blocking electrodes. Data were collected in a frequency range from 0.01 Hz to 1 MHz using an electrochemical workstation (IM6, Germany) from room temperature to 200 °C. Finally, the data were analyzed and fitted by customized equivalent circuits using ZView. The galvanostatic cycling of the all-solid-state battery was tested using MACCOR in the voltage range of 0.8–4.1 V at 0.2 °C at room temperature.

Results and discussion

Figure 1 shows the DSC thermograms of $\text{Li}_{1.5}\beta_x\text{Al}_{0.5-x}\text{Ge}_{1.5}(\text{PO}_4)_3$. The glass transition temperatures, T_g , crystallization temperatures, T_c , and the difference between T_c and T_g are tabulated in Table 1. It can be seen that when $\beta = \text{B}$, In , T_c , and T_g increased with the increase in the amount of dopants. The T_c of the pure LAGP is 614 °C, while for all B- and In-doped LAGP, their T_c values are higher. By contrast, the T_c

and T_g of the Ga-doped samples decrease with the increase in the content of Ga, implying that Ga promotes the crystallization of LAGP.

Figure 2 shows the XRD spectra for B-, Ga-, and In-doped LAGP. All of the XRD spectra show the main phase of NASICON with some weak peaks that could be assigned to AlPO_4 and GeO_2 [23, 24, 28, 29]. No peaks corresponding to precursors of B_2O_3 , Ga_2O_3 , and In_2O_3 were observed, implying in cooperation of those doped elements in the LAGP lattices. Compared with the three doped LAGP, it is noted that for the In-doped LAGP, obvious diffraction peaks of AlPO_4 near $2\theta = 13^\circ$ and 16° could be observed (Fig. 2c) which may indicate that AlPO_4 has lower Gibbs energy. As the ionic radii of B^{3+} , Al^{3+} , Ga^{3+} , and In^{3+} are 27 pm, 53.5 pm, 62 pm, and 80 pm, respectively, with the increase of the ionic radius, the interaction among different ionic increased. Considering that Al^{3+} owns the strongest reduction among these ions ($\text{Al}^{3+} > \text{B}^{3+}$ and $\text{Al}^{3+} > \text{Ga}^{3+} > \text{In}^{3+}$), Al^{3+} would easily separate out from the structure as AlPO_4 .

Figure 3 shows the electrochemical impedance spectra (EIS) of LA β GP measured at room temperature. Depressed semicircles were observed for all the specimens. The intercept of the semicircle with the real axis at the high frequency corresponds to the bulk resistance, R_b , and the diameter of the semicircle corresponds to the resistance at the grain boundary, R_g . The total resistance, R_t , of a specimen is the sum of the bulk resistance, R_b , and grain boundary resistance, R_g . The semicircle was followed by Warburg-type impedance at low frequencies, which is associated with the diffusion process. The total resistance ($R_t = R_b + R_g$) was normalized to the thickness and blocking electrode area of each specimen to obtain the ionic conductivity, σ_t . The ionic conductivities of all the specimens were calculated using the equation that listed in the reference [30]. For the B-doped LAGP ($\text{Li}_{1.5}\text{Al}_{0.3}\text{B}_{0.2}\text{Ge}_{1.5}(\text{PO}_4)_3$), a maximum total ionic conductivity of 5.26×10^{-4} S/cm was achieved, which may result from the well-defined and compact connections among the grains. With the further increase in the content of B_2O_3 , the total ionic conductivity decreased, which may result from the increase of the concentration of AlPO_4 in the grain boundary. For Ga-doped LAGP, the highest bulk ionic conductivity of 6.67×10^{-4} S/cm was obtained at $x = 0.1$. It is also found that the highest total ionic conductivity of LABGP ($x = 0.2$) sample is more than 2 times higher than the reported value of

Table 1 Glass transition temperature and crystallization temperature of $\text{Li}_{1.5}\beta_x\text{Al}_{0.5-x}\text{Ge}_{1.5}(\text{PO}_4)_3$ ($\beta = \text{B}$, Ga , In , $x = 0.1\sim 0.5$)

β (ionic radius/Å)	x	T_g (°C)	T_c (°C)	$T_c - T_g$ (°C)
B (0.27)	0.1	526	623	97
	0.2	530	624	94
	0.3	538	626	88
	0.4	535	630	95
	0.5	561	633	72
Al (0.535)	0	501	614	113
Ga (0.62)	0.1	516	620	104
	0.2	512	612	100
	0.3	510	609	99
	0.4	506	602	96
	0.5	504	599	95
In (0.8)	0.1	525	626	101
	0.2	531	647	116
	0.3	535	664	129
	0.4	541	704	163
	0.5	542	702	160

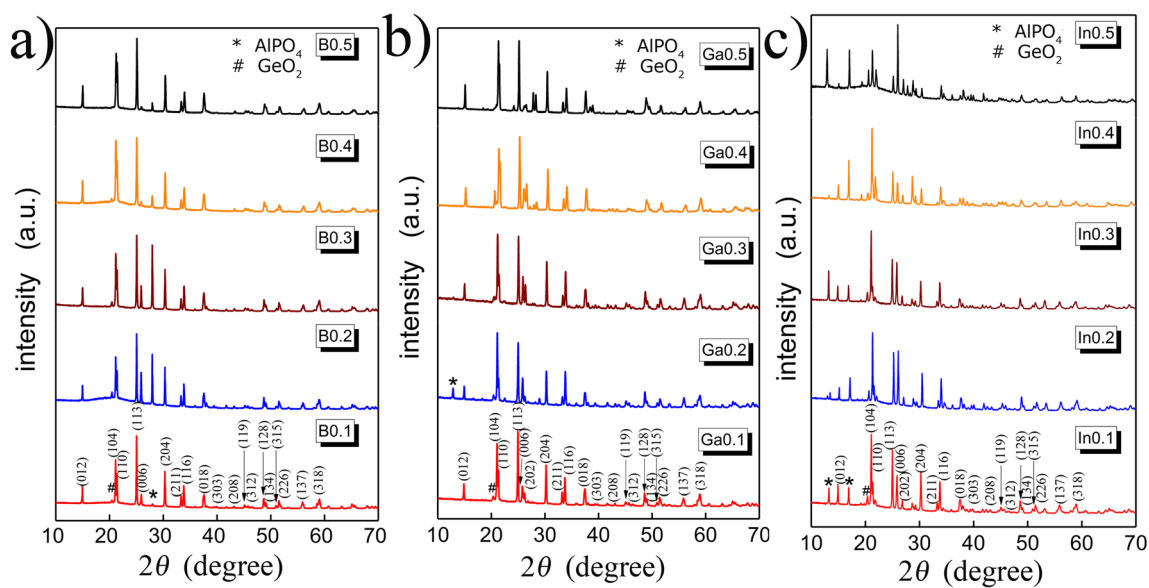


Fig. 2 XRD results of $\text{Li}_{1.5}\text{Al}_{0.5-x}\beta_x\text{Ge}_{1.5}(\text{PO}_4)_3$, $x = 0.1 \sim 0.5$: (a) $\beta = \text{B}$, (b) $\beta = \text{Ga}$, (c) $\beta = \text{In}$

Fig. 3 EIS results of $\text{Li}_{1.5}\text{Al}_{0.5-x}\beta_x\text{Ge}_{1.5}(\text{PO}_4)_3$: (a) and (b) $\beta = \text{B}$, (c, d) $\beta = \text{Ga}$, and (e, f) $\beta = \text{In}$

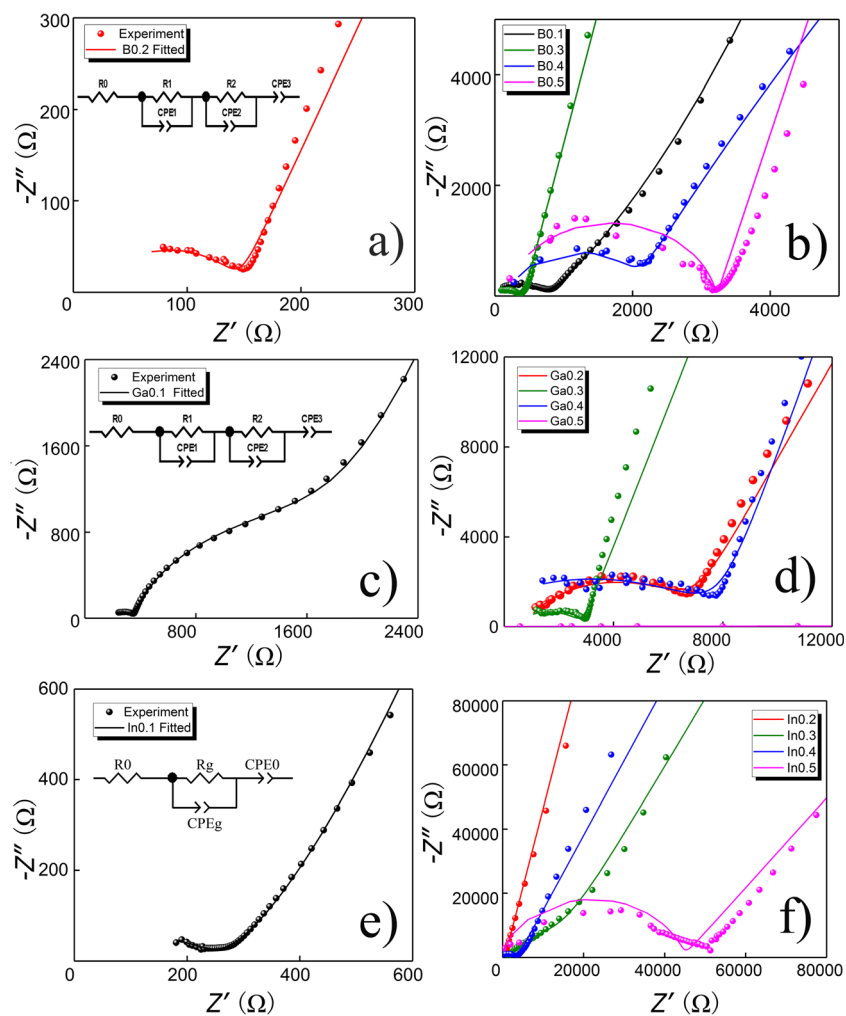
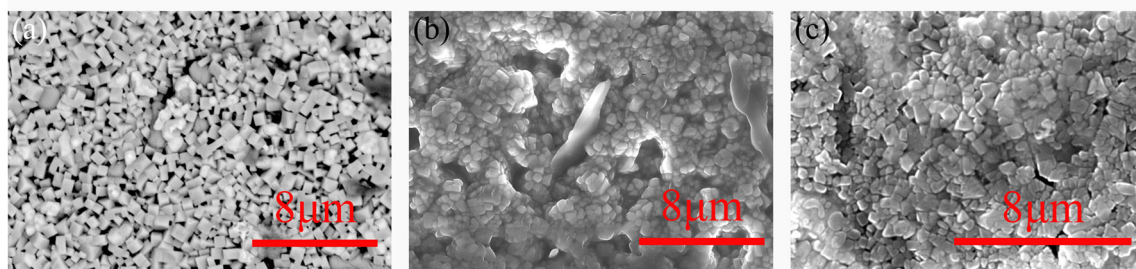
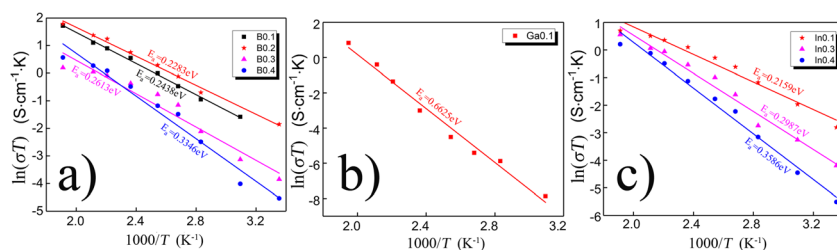
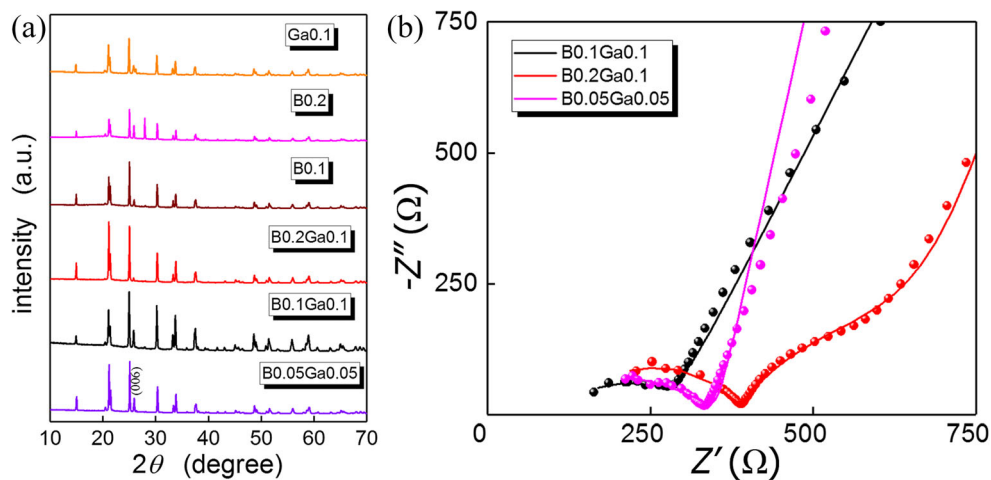


Table 2 Ionic conductivity at room temperature and activation energy of various components

β	X	Ionic conductivity (σ_i , 10^{-4} S/cm, RT)	Activation energy (eV)	Density (g/cm ³)	Statistical errors of σ_i
B	0.1	0.789 (total)	0.2438	4.7749	0.99737
	0.2	5.26 (total)	0.2283		0.99368
	0.3	0.109 (total)	0.2613		0.94968
	0.4	0.132 (total)	0.3346		0.9713
	0.5	0.36 (total)			
Ga	0.1	6.67 (bulk)	0.6625	3.42694	0.98481
	0.2	1.67 (total)			
	0.3	3.47 (total)			
	0.4	1.02 (total)			
	0.5	5.79 (bulk)			
In	0.1	2.11 (total)	0.2159	3.68182	0.96927
	0.2	1.54 (total)			
	0.3	0.506 (total)	0.2987		0.9787
	0.4	0.136 (total)	0.3586		0.98649
	0.5	0.0144 (total)			
B0.05Ga0.05		1.83 (total)		3.87294	0.98759
B0.1Ga0.1		3.86 (total)	0.2495		
B0.2Ga0.1		2.43 (bulk)			

Fig. 4 Arrhenius plot of $\text{Li}_{1.5}\text{Al}_{0.5-x}\beta\text{Ge}_{1.5}(\text{PO}_4)_3$: (a) $\beta = \text{B}$, (b) $\beta = \text{Ga}$, and (c) $\beta = \text{In}$ **Fig. 5** SEM images: (a) singly doped by Ga0.1, (b) singly doped B0.1, and (c) co-doped B0.1 and Ga0.1**Fig. 6** (a) XRD and (b) EIS results of B and Ga doping samples

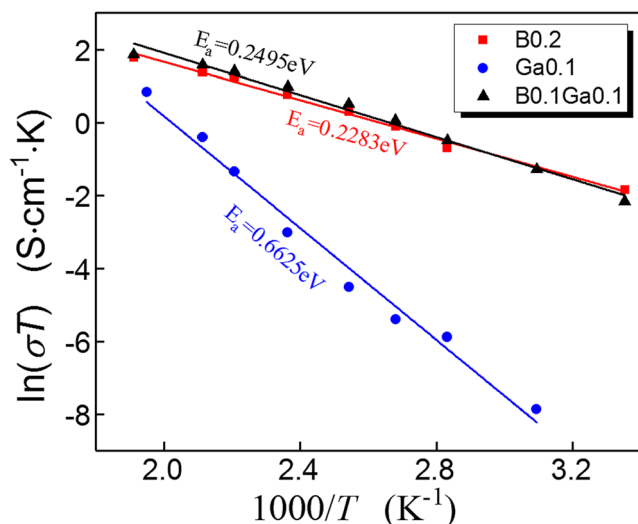


Fig. 7 Arrhenius plot of B0.1, Ga0.1, and B0.1Ga0.1

the pure LAGP (2.4×10^{-4} S/cm) [31], whereas the total ionic conductivity of the In-doped samples decreased with the increase in In content. The decrease in the ionic conductivity of the In-doped LAGP is caused by the formation of the secondary phase according to XRD results due to much bigger size of In^{3+} than that of Al^{3+} . The maximum total ionic conductivity of the In-doped specimens is 2.11×10^{-4} S cm^{-1} when $x = 0.1$ ($\text{Li}_{1.5}\text{Al}_{0.4}\text{In}_{0.1}\text{Ge}_{1.5}(\text{PO}_4)_3$). Table 2 summarizes the ionic conductivities of all specimens.

Figure 4 shows the $\ln(\sigma T) - 1000/T$ plot. The slope of the fitted line can be used to calculate the activation energy of the solid electrolyte based on Eq. (1) [30].

$$E_a = -R \frac{d \ln(\sigma T)}{d(1000/T)} \quad (1)$$

where E_a is the activation energy, R is the gas constant ($8.3144 \text{ J mol}^{-1} \text{ K}^{-1}$), and $\frac{d \ln(\sigma T)}{d(1000/T)}$ equals to the slope of the fitted line in the $\ln(\sigma T) - 1000/T$ plot. It can be seen from Fig. 4 and Table 2 that the activation energy of B-doped samples is significantly lower than that of the pure LAGP (0.38 eV) and the Ga-doped samples [31]. When $x = 0.2$, the lowest activation energy of $E_a = 0.2283$ eV was obtained. For In-doped samples, a comparable activation energy to LABGP ($x = 0.2$)

was obtained at $x = 0.1$. The low activation energy indicates less sensitive to temperature change.

Figure 5 shows the morphology of the cross-section of B- and Ga-doped samples. It can be seen from Fig. 5a) that, for Ga-doped sample, the shape of grains is regular cubes with sharp edges and the grain size distribution is uniform, indicating good crystallization. There exist some spaces between grains. In contrary, for B-doped sample as shown in Fig. 5b, the edges of grains were smooth and unclear. The grains were tightly glued together by the gel-like grain boundary and no space between grains was observed. For B0.1- and Ga0.1-co-doped LAGP as shown in Fig. 5c, the grains are closely packed and the size of grains is uniform although the shape of grains is irregular. The results are consistent with the density as shown in Table 2 that B0.2-doped LAGP owns a higher density as much as 4.7749 than B0.2- and Ga0.1-co-doped LAGP as much as 3.87294.

Since both B and Ga can largely improve ionic conductivity of LAGP, B and Ga were used to co-dope to LAGP. Figure 6 a shows the XRD patterns of the B- and Ga-co-doped LAGP electrolytes. No impurity peaks can be observed, which indicate that B and Ga were fully incorporated into NASICON structure. Figure 6 b shows the EIS of the B- and Ga-co-doped LAGP at room temperature. The ionic conductivities are shown in Table 2. It can be seen that when the content of Ga is 0.1, the total ionic conductivity decreased with the increase of boron. The maximum total ionic conductivity is 3.86×10^{-4} S/cm when $B = 0.1$ and $Ga = 0.1$ which is higher than that of B0.1, however, a little lower than B0.2.

Figure 7 shows the Arrhenius plot of B0.1, Ga0.1, and B0.1Ga0.1. For B- and Ga-co-doped samples, LAGP-B0.1Ga0.1 exhibited an activation energy as low as 0.2495 eV ($23.9864 \text{ kJ mol}^{-1}$) which was much lower than Ga-LAGP ($x = 0.1$), and the reported results of LAGP (0.6625 eV) that may indicate boron can effectively reduce activation energy [28–30].

The charge-discharge performance of LNMO/ $\text{Li}_{1.5}\text{Al}_{0.4}\text{Ga}_{0.1}\text{Ge}_{1.5}(\text{PO}_4)_3/\text{RuO}_2$ full cell was measured at room temperature, and the results are shown in Fig. 8. It can be seen that the coulombic efficiency increased with the cycles and up to about 80% after 40 cycles which should result from the improvement of interfaces between the electrolyte and the electrodes. The

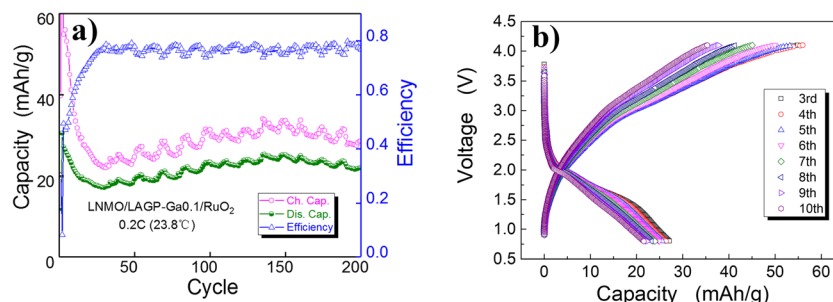
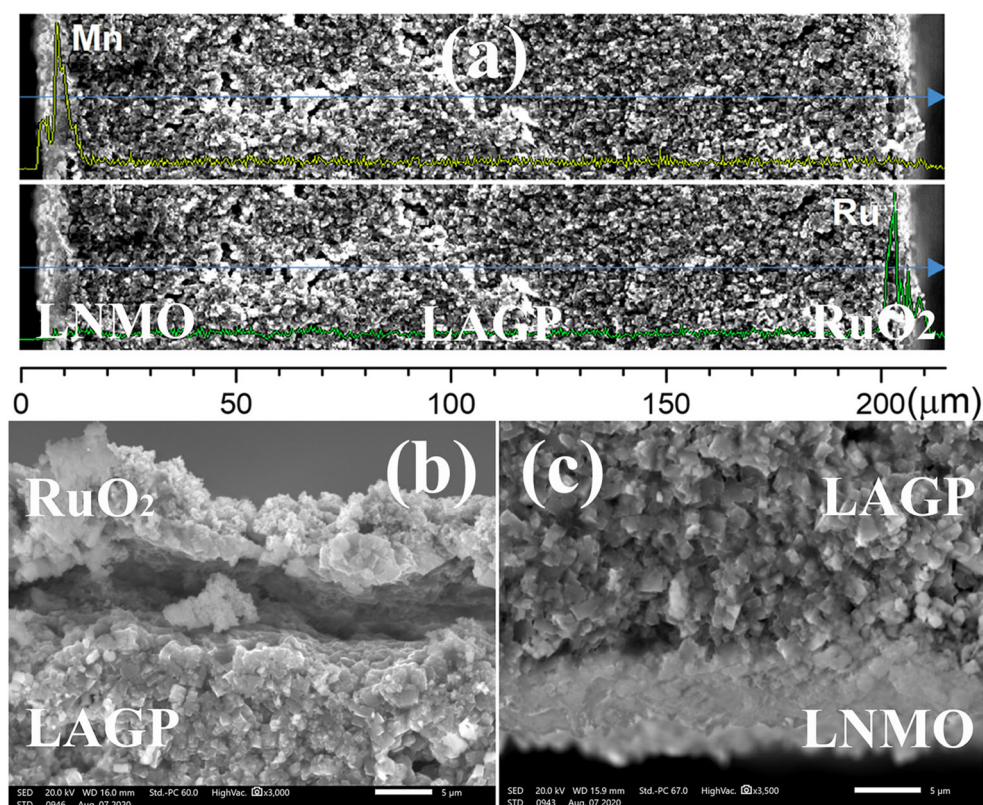


Fig. 8 (a) Charge capacity, discharge capacity, and efficiency of LNMO/ $\text{Li}_{1.5}\text{Al}_{0.4}\text{Ga}_{0.1}\text{Ge}_{1.5}(\text{PO}_4)_3/\text{RuO}_2$ and (b) voltage-capacity curve of 3rd–10th cycles

Fig. 9 Cross-section of the all-solid-state battery after 200 cycles: **(a)** whole cross-section with edx peaks of Mn and Ru, **(b)** RuO_2 /LAGP interface, **(c)** LNMO/LAGP interface



capacity of charge and discharge decreased with the increase of cycles and tended to be stable after 40 cycles and showed a stable discharge capacity as much as ~ 20 mAh/g. Because of high internal impedance and poor activation of the interface, the polarization of this cell was obvious as shown in Supplementary Fig. S1. As a result, oscillations occurred in the first charge process, and the initial voltage dropped sharply to 2.75 V after jumping to 4.1 V. The second discharge specific capacity was 30 mAh/g with a low efficiency of 8.2%. This low coulombic efficiency in the first 2 cycles may indicate the activation of the electrode/electrolyte interface. For the rest cycles, the charge plateau (3.0–4.1 V) and discharge plateau (3.0–0.8 V) which were corresponding to the oxidation and reduction peaks of the electrode materials can be clarified that proving the ASSLiB was successfully operated based on microwave-sintered $\text{LA}\beta\text{GP}$. The low capacity may result from the performance of interface [31] and hydrogen that introduced during assembling full battery [32], and further studies are in progress and will be reported in future work.

Figure 9 shows the SEM image of the full battery after 200 charge-discharge cycles. As can be seen from Fig. 9a, the thickness of cathode and anode is about 5 μm, and the EDX peaks of Mn and Ru can only be separately observed in cathode and anode which indicate that no obvious diffusion occurred in electrode/electrolyte interfaces. For anode/electrolyte interface, as shown in Fig. 9b, RuO_2 seemed to space out from electrolyte that may result from the volume

change of RuO_2 . For cathode/electrolyte interface, as shown in Fig. 9c, the cathode layer is dense and well contacted with the electrolyte which remained good crystallization, and no obvious voids could be found in the interface between $\text{LA}\beta\text{GP}$ and the cathode layer which implies that the cathode/electrolyte interface is stable in charge-discharge cycles. In a word, microwave-sintered $\text{LA}\beta\text{GP}$ can be successfully used in all-solid-state lithium-ion batteries by the help of PAALi, but the anode/electrolyte need to be optimized.

Conclusion

In summary, $\text{Li}_{1.5}\text{Al}_{0.5-x}\beta_x\text{Ge}_{1.5}(\text{PO}_4)_3$ ($\beta = \text{B, Ga, In}$, $x = 0.1\sim 0.5$) have been successfully prepared by microwave-assisted crystallization process. A maximum total ionic conductivity of 5.26×10^{-4} S/cm was obtained when $\beta = \text{B}$, $x = 0.2$. In cross-sectional morphology of $\text{LA}\beta\text{GP}$ ($x = 0.1$), crystal grains touched well each other, and no space between the grains was observed, leading to improve grain boundary conductivity. When B and Ga were co-doped, a maximum total ionic conductivity of 3.86×10^{-4} S/cm was obtained when $B = 0.1$ and $Ga = 0.1$. Although the total ionic conductivity is about the same as that of the pure LAGP , the activation energy is significantly improved. An activation energy as low as 0.2495 eV was obtained. It can be speculated that Ga

can improve the bulk ionic conductivity and B can reduce the activation energy.

Funding This work was supported by the National Natural Science Foundation of China [51475174], Xiamen Industry-University-Research Collaborative Innovation Project [3502Z20193026], Scientific Research Funds of Huaqiao University [Z16Y0020], and Subsidized Project for Postgraduates' Innovative Fund in Scientific Research of Huaqiao University.

References

- Kang K, Meng YS, Bréger J, Grey CP, Ceder G (2006) Electrodes with high power and high capacity for rechargeable lithium batteries. *Science* 311(5763):977–980
- Armand M, Tarascon JM (2008) Building better batteries. *Nature* 451(7179):652–657
- Sun W, Wang Y (2014) Graphene-based nanocomposite anodes for lithium-ion batteries. *Nanoscale* 6(20):11528–11552
- Etacheri V, Marom R, Ran E, Salitra G, Aurbach D (2011) Challenges in the development of advanced Li-ion batteries: a review. *Energy Environ Sci* 4(9):3243–3262
- Scrosati B, Garche J (2010) Lithium batteries: status, prospects and future. *J Power Sources* 195(9):2419–2430
- Nagai H, Hara H, Enomoto M, Mochizuki C, Honda T, Takano I, Sato M (2013) Synchronous electrochromism of lithium ion battery with chemically fabricated transparent thin films. *Funct Mater Lett* 06(02):1341001
- He L, Chen C, Kotobuki M, Zheng F, Zhou H, Lu L (2019) A new approach for synthesizing bulk-type all-solid-state lithium-ion batteries. *J Mater Chem A* 7(16):9748–9760
- Goodenough JB, Hong YP, Kafalas JA (1976) Fast Na⁺ ion transport in skeleton structures ☆. *Mater Res Bull* 11(2):203–220
- Hong YP (1976) Crystal structures and crystal chemistry in the system Na_{1+x}Zr₂Si_xP_{3-x}O₁₂ ☆. *Mater Res Bull* 11(2):173–182
- Liu Y, Li C, Li B, Song H, Cheng Z, Chen M, He P, Zhou H (2018) Germanium thin film protected lithium aluminum germanium phosphate for solid-state Li batteries. *Adv Energy Mater*:1702374
- Bachman JC, Muy S, Grimaud A, Chang HH, Pour N, Lux SF, Paschos O, Maglia F, Lupart S, Lamp P (2016) Inorganic solid-state electrolytes for lithium batteries: mechanisms and properties governing ion conduction. *Chem Rev* 116(1):140–162
- Knauth P (2009) Inorganic solid Li ion conductors: an overview. *Solid State Ionics* 180(14):911–916
- Wang Y, Richards WD, Ong SP, Miara LJ, Kim JC, Mo Y, Ceder G (2015) Design principles for solid-state lithium superionic conductors. *Nat Mater* 14(10):1026–1031
- Zheng F, Kotobuki M, Song S, Lai MO, Lu L (2018) Review on solid electrolytes for all-solid-state lithium-ion batteries. *J Power Sources* 389:198–213
- Song S, Duong HM, Korsunsky AM, Hu N, Lu L (2016) A Na(+) superionic conductor for room-temperature sodium batteries. *Sci Rep* 6(1):32330
- Song S, Kotobuki M, Zheng F, Xu C, Savilov SV, Hu N, Lu L, Wang Y, Li WZ (2017) A hybrid polymer/oxide/ionic-liquid solid electrolyte for Na-metal batteries. *J Mater Chem A* 5(14):6424–6431
- Xu X, Wen Z, Wu X, Yang X, Gu Z (2007) Lithium ion-conducting glass-ceramics of Li_{1.5}Al_{0.5}Ge_{1.5}(PO₄)₃-xLi₂O (x=0.0–0.20) with good electrical and electrochemical properties. *J Am Ceram Soc* 90(9):2802–2806
- Aono H, Imanaka N, Adachi GY (1994) High Li⁺ conducting ceramics. *Acc Chem Res* 27(9):265–270
- He K, Zu C, Wang Y, Han B, Yin X, Zhao H, Liu Y, Chen J (2014) Stability of lithium ion conductor NASICON structure glass ceramic in acid and alkaline aqueous solution. *Solid State Ionics* 254(4):78–81
- CRETIN (1999) Comparative study of lithium ion conductors in the system Li_{1+x}Al_xA_{2-x}IV (PO₄)₃ with AIV = Ti or Ge and 0 ≤ x ≤ 0.7 for use as Li⁺ sensitive membranes. *J Eur Ceram Soc* 19(16):2931–2940
- Mariappan CR, Yada C, Rosciano F, Roling B (2011) Correlation between micro-structural properties and ionic conductivity of Li_{1.5}Al_{0.5}Ge_{1.5}(PO₄)₃ ceramics. *J Power Sources* 196(15):6456–6464
- Illbeigi M, Fazlali A, Kazazi M, Mohammadi AH (2016) Effect of simultaneous addition of aluminum and chromium on the lithium ionic conductivity of LiGe₂(PO₄)₃ NASICON-type glass-ceramics. *Solid State Ionics* 289:180–187
- Chowdari BVR, Kumari PP (1996) Synthesis and characterization of silver borotellurite glasses. *Solid State Ionics* 86(96):521–526
- Fan JL, Huang BY, Liu J, En-Xi WU (2004) Principles and status of microwave sintering. *Powder Metall Ind* 14(1):29–33
- Kothari DH, Kanchan DK (2016) Effect of doping of trivalent cations Ga³⁺, Sc³⁺, Y³⁺ in Li_{1.5}Al_{0.5}Ge_{1.5}(PO₄)₃ (LATP) system on Li⁺ ion conductivity. *Phys B* 501:90–94
- Kotobuki M, Hanc E, Yan B et al (2017) Preparation and characterization of Ba-substituted Li_{1+x}Al_xGe_{2-x}(PO₄)₃ (x = 0.5) solid electrolyte. *Ceram Int* 43(15):12616–12622
- Zhu Y, Zhang Y, Lu L (2015) Influence of crystallization temperature on ionic conductivity of lithium aluminum germanium phosphate glass-ceramic. *J Power Sources* 290:123–129
- Sung Hwa J, Jin-Ho L, Forster PM, Gérard F, Cheetham AK, Jong-San C (2006) Microwave synthesis of hybrid inorganic-organic porous materials: phase-selective and rapid crystallization. *Chem Eur J* 12(30):7899–7905
- Fu J (2000) Effects of M³⁺ ions on the conductivity of glasses and glass-ceramics in the system Li₂O—M₂O₃—GeO₂—P₂O₅ (M = Al, Ga, Y, Dy, Gd, and La). *J Am Ceram Soc* 83(4):1004–1006
- Martínez-Juárez A, Pecharrromán C, Iglesias JE, Rojo JM (1998) Relationship between activation energy and bottleneck size for Li⁺ ion conduction in NASICON materials of composition LiMM'(PO₄)₃; M, M' = Ge, Ti, Sn, Hf. *J Phys Chem B* 102(2):372–375
- Yu S, Mertens A, Tempel H, Schierholz R (2018) Monolithic all-phosphate solid-state lithium-ion battery with improved interfacial compatibility. *ACS Appl Mater Interfaces* 10(26):22264–22277
- Morita K, Tsuchiya B, Tsuchiya H, Majima T (2020) Change in Li depth profiles of Au/LCO/mixed LATP-LAGP/Pt battery under discharging studied by ion beam analysis techniques with 9MeV O⁴⁺ ions. *Solid State Ionics* 344:115–135

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